

Noether's theorem

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Let B be a real Poisson algebra, and write $\text{ad}_f = \{f, -\}$ for $f \in B$. The map $\text{ad} : B \rightarrow \text{ad}(B)$ is a Lie algebra homomorphism onto the Lie algebra of Hamiltonian derivations, with kernel the Poisson centre Z .

Let D be a Poisson derivation of B , so that $D(Z) \subseteq Z$. Write $B^D = \ker D$, $Z^D = Z \cap B^D$, $DZ = D(Z)$ and $\text{ad}(B)^D = \{Y \in \text{ad}(B) : [D, Y] = 0\}$.

Lemma 1. *There is an exact sequence of Lie algebras*

$$0 \longrightarrow Z^D \longrightarrow B^D \xrightarrow{\text{ad}} \text{ad}(B)^D \xrightarrow{\partial} Z/DZ,$$

where Z/DZ is abelian and $\partial(\text{ad}_f) = [Df]$.

Proof. The identity $[D, \text{ad}_f] = \text{ad}_{Df}$ shows that $\text{ad}_f \in \text{ad}(B)^D$ if and only if $Df \in Z$. Since $\ker \text{ad} = Z$, the class $[Df]$ depends only on ad_f , so ∂ is well defined. If $Df, Dg \in Z$, then

$$D\{f, g\} = \{Df, g\} + \{f, Dg\} = 0,$$

so ∂ is a Lie algebra homomorphism. Finally, $\partial(\text{ad}_f) = 0$ if and only if $f \in B^D + Z$. It follows that $\ker \partial = \text{im}(\text{ad}|_{B^D})$, while $\ker(\text{ad}|_{B^D}) = Z^D$. ■

Now let (M, ω) be a non-empty connected symplectic manifold, let $I \subset \mathbb{R}$ be an open interval containing 0, and let $H \in C^\infty(M)$. Give $A = C^\infty(M \times I)$ its fibrewise Poisson bracket. For $f \in A$, define the vertical field X_f by $\iota_{X_f}\omega = d_M f$, so that $X_f(g) = \{g, f\}$ and $\text{ad}_f = -X_f$. The operator $D = \partial_t + X_H$ is a Poisson derivation of A .

Theorem 2 (Noether). *There is a central extension of Lie algebras*

$$0 \longrightarrow \mathbb{R} \longrightarrow A^D \xrightarrow{\text{ad}} \text{ad}(A)^D \longrightarrow 0.$$

Proof. Non-degeneracy of ω and connectedness of M give $Z(A) = C^\infty(I)$. On this centre, D is ordinary differentiation, which is surjective and has kernel $\mathbb{R}$. Apply Lemma 1. ■

Remark. If $\text{ad}_f \in \text{ad}(A)^D$, then $Df \in Z(A)$, but Df need not vanish. Choose $h \in Z(A)$ with $Dh = Df$. Since $D = \partial_t$ on $Z(A) = C^\infty(I)$, one such choice is

$$h(t) = \int_0^t (Df)(s) ds.$$

Then $F = f - h$ satisfies $\text{ad}_F = \text{ad}_f$ and $DF = 0$. Along any solution $\dot{z}(t) = X_H(z(t))$,

$$\frac{d}{dt}F(z(t), t) = (DF)(z(t), t) = 0.$$

Thus F is conserved.

Example 3. Consider a particle of mass $m > 0$ in a uniform gravitational field, with x measured upwards. On $M = \mathbb{R}_x \times \mathbb{R}_p$, take $\{x, p\} = 1$ and

$$H = \frac{p^2}{2m} + mgx,$$

where $g > 0$. A spatial translation changes H only by a constant and leaves its Hamiltonian equations unchanged. Here

$$D = \partial_t + \frac{p}{m} \partial_x - mg \partial_p.$$

The derivation $\text{ad}_p = -\partial_x$ commutes with D , but $Dp = -mg$. Taking $h = -mgt$ gives $F = p + mgt$, so $p(t) + mgt = p(0)$ along every trajectory.